

MD. AHASHAN HABIB

AI/ML Engineer

📍 Dhaka, Bangladesh 📞 (+880) 1709-180782 ✉ mr.ahashan261@gmail.com
🌐 github.com/mr-ahabib 🔗 linkedin.com/in/md-ahashan-habib 🌐 mr-ahabib.github.io

PROFESSIONAL SUMMARY

AI/ML Engineer and Researcher with expertise in Large Language Models, OCR, RAG systems, and Deep Learning. Proven track record in developing production-grade AI solutions for education and enterprise. Published researcher with **3 IEEE publications** and **2 papers under review**. Strong background in Computer Science with hands-on experience in AI-powered document processing, knowledge base systems, and intelligent tutoring platforms.

PROFESSIONAL EXPERIENCE

AI/ML Engineer (Contractual)
Qanun Limited, Dhaka, Bangladesh

Feb 2026 – Apr 2026

- Developed AI-powered OCR and intelligent document processing solutions for multilingual scanned documents in Bangla, Arabic, English, and mathematical notations
- Built document digitization pipelines to convert scanned files into searchable, structured, and AI-ready knowledge bases for enterprises
- Developed RAG-based enterprise knowledge base systems with intelligent document retrieval and AI chatbot integration for faster information access

AI/ML Engineer
Ethics Advance Technology Ltd., Dhaka, Bangladesh

Feb 2025 – Jan 2026

- Developed and deployed AI-powered Medical Education Platform serving 800+ medical students with intelligent tutoring
- Applied domain-adapted LLMs with RAG and vector databases for intelligent content delivery and assessment generation
- Built production-grade LLM solutions integrated with web-based LMS for real-time educational assistance
- Ensured responsible AI practices including fairness, bias mitigation, and ethical AI considerations
- Collaborated with medical professionals to develop AI-driven educational content and automated assessment tools

Research Assistant
LUT University, Finland (Remote)

Aug 2024 – Jan 2025

- Conducted research on reinforcement learning-based knowledge editing techniques for Large Language Models to enable efficient factual knowledge updates
- Developed methods to identify optimal transformer layers and edit directions using RL strategies for targeted memory modification
- Implemented parameter-efficient knowledge editing approaches focusing on edit efficacy, specificity, and knowledge retention in GPT-based architectures
- Worked on model merging and ranking techniques to evaluate factual knowledge across multiple transformer-based language models

TECHNICAL SKILLS

AI/ML: LLMs, NLP, Deep Learning, RAG, PyTorch, TensorFlow, Transformers, Hugging Face, LangChain, Scikit-learn

Programming: Python, Java, C, C++, JavaScript, TypeScript, Solidity

Frameworks: FastAPI, Django, Node.js, React, React Native, Express.js

Databases: MySQL, PostgreSQL, FAISS, Vector Databases, PgVector

Specialized: RAG, OCR, Computer Vision, Blockchain, Document Processing, Knowledge Base Systems

DevOps & Tools: Docker, Git, CI/CD, Linux, HPC, REST APIs, JWT, OAuth2, LUMI Super Computing

EDUCATION

Bachelor of Science in Computer Science and Engineering
United International University, Dhaka, Bangladesh

Jan 2020 – Dec 2024
CGPA: 3.62/4.00

- **Thesis:** A Secure Blockchain Based Brain Tumor Prediction By Using Swin Transformer
- **Relevant Coursework:** Machine Learning, Deep Learning, NLP, Data Structures, Algorithms, Database Systems

Higher Secondary Certificate (HSC)
Govt. Azizul Haque College, Bogra, Bangladesh

Jul 2017 – Apr 2019
GPA: 4.83/5.00

KEY PROJECTS

- **AI-Powered Medical Education Platform** | LLMs, RAG, FastAPI, React | ai-lms.eatlbd.com
 - Production LMS serving 800+ students with AI-driven content delivery and automated assessment generation
 - Implemented domain-adapted LLMs with RAG for personalized learning and semantic content retrieval
- **DoceanAI Cloud Platform** | FastAPI, LLM, OCR, Knowledge Base, React | doceanai.cloud
 - Cloud-based AI platform for intelligent document processing and analysis with production-grade deployment
 - Developed scalable AI solutions for document understanding, OCR, and natural language processing
- **Bangla PDF RAG System** | FastAPI, RAG, FAISS, OCR, React | [GitHub](https://github.com)
 - Intelligent system combining OCR, semantic search, and RAG for Bangla PDF question-answering
- **DocLink Medical App** | Express.js, React Native, TypeScript, MySQL, OCR | [GitHub](https://github.com)
 - Healthcare application with symptom-based disease classification and health record management
- See more projects on [GitHub](https://github.com)

PUBLICATIONS

- **Habib, M. A.**, Sarwar, M. W., Hasan, M., & Mokaddis, M. A. (2025, October). A Secure Blockchain Based Brain Tumor Prediction Model Using Swin Transformers. In 2025 16th IEEE International Conference on Industry Applications (INDUSCON) (pp. 1933-1940). IEEE, Brazil. [\[First Author\]](#)
- **Habib, M. A.**, Alvi, R., Sarkar, S., Shuvo, M. A., Hasan, M., & Azim, R. (2025, July). ADDomics: Enhancing Multiomics Fusion with Autoencoders and Deep Neural Networks. In 2025 IEEE Region 10 Symposium (TENSYP) (pp. 1-8). IEEE, New Zealand. [\[First Author\]](#)
- Sifat, S. A., Hasan, M., Joy, S. B. N., **Habib, M. A.**, & Rahman, R. (2024, May). PyroVision: A deep learning based model for wildfire detection in satellite imagery. In 2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT) (pp. 705-710). IEEE, Dhaka, Bangladesh. [\[Co-Author\]](#)

Under Review

- RL-MR: Reinforcement Learning for Dynamic Pretrained Model Ranking. *Under Review at EMNLP.* [\[First Author\]](#)
- ReinKEdit: A Reinforcement Learning for Knowledge Editing in Large Language Models. *Under Review at Pattern Recognition.* [\[First Author\]](#)

HONORS AND AWARDS

- **Reviewer, IEEE Region 10 Symposium (TENSYP)** – Peer reviewer for AI and ML research papers 2025
- **Champion, Kaggle Competition** – First place in university-level data science competition 2024
- **Champion, Software Lab CSE Project Show Fall '24** – Best project award for innovative solution 2024
- **Bangladesh Scouts Award** – National award for leadership and community service from Prime Minister 2012

REFERENCES

Dr. Saddam Hossain Mukta
Associate Professor
Department of Computer Science & Engineering
LUT University, Finland
Email: saddam.mukta@lut.fi

Dr. Riasat Azim
Associate Professor
Department of Computer Science & Engineering
United International University, Bangladesh
Email: riasat@cse.uiu.ac.bd